

Darius Farngalo

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EDUCATION

University at Buffalo, The State University of New York

Buffalo, NY

Bachelor of Science in Computer Science

Expected Graduation, May 2025

Relevant Courses: Algorithms and Complexity, Data Structures, Object Oriented Programming, Systems Programming, Intro to Artificial Intelligence, Intro to Probability & Statistics, Web Applications, Linear Algebra

TECHNICAL SKILLS

Languages: Python, Java, C (Programming Language), C++, C#, Go, MIPS Assembly, PowerShell, Javascript, HTML/CSS

Technologies and Tools: PostgreSQL, SQL, MongoDB, Jupyter Notebook, ROS, Unreal Engine, Conda, OpenAI, Git, GitHub, GDB, Emacs, Linux, Unix, MATLAB, Microsoft Office

Frameworks: React, Flask, Django, RESTful API, AWS, Docker, JUnit

EXPERIENCE

Google (G-SWEP) via ProjectBasta

Remote

Software Engineer Apprentice

July 2024 - Present

- Selected as one of **80 participants** out of **350+ applicants** for an intensive 10-week mentorship program led by Google Software Engineers, focused on enhancing technical problem-solving and interview skills.
- Engaged in weekly 1:1 sessions with a Google SWE mentor, receiving personalized guidance on data structures, algorithms, and system design.

The Duhart Group

Remote

Software Engineer Intern

June 2024 - Present

- Designed and implemented an **MLOps pipeline** leveraging machine learning for automated data extraction and processing, ensuring scalability and reliability for autonomous systems using **AWS S3, Lambda, and RDS**.
- Implemented a robust **MLOps pipeline** with **Amazon SageMaker** and **GitHub Actions** to streamline model deployment, enhancing scalability and reliability by **63%**.
- Implemented **CI/CD practices** by automating testing and deployment processes using **Git** and **GitHub Actions**, ensuring seamless integration and continuous delivery of updates.

Unfold Studios

Buffalo, NY

AI Research Intern

June 2024 - August 2024

- Integrated advanced AI technologies, including Large Language Models (**LLMs**) and natural language processing (**NLP**), to enhance interactive storytelling and create user-centric educational tools.
- Utilized **Python** to develop and optimize machine learning algorithms for data training and prediction in autonomous systems, focusing on real-time decision-making and scalability.
- Utilized **Django** and **PostgreSQL** to manage data from **15,000+ users**, improving personalized content delivery, scaling the platform, and boosting user retention and learning outcomes.

TECHNICAL PROJECTS

Full-Stack Fitness and Wellness Web Application

- Led the development of a full-stack fitness web application, enhancing physical and mental health with integrated features like a personal journal and a DIY meditation section.
- Utilized **Docker, JavaScript, Python, Flask, and Bash** for robust server-side functionalities and effective **MongoDB** database management, promoting user engagement and ensuring data integrity without relying on pre-existing frameworks.

Media Library Data Management System

- Engineered a media library system in **Java** to manage and analyze **4,575 movies** and **2,000 songs**, utilizing advanced data structures (linked lists, binary search trees, hash maps) to enhance data processing efficiency by **50%**.
- Implemented file reading and data parsing algorithms for importing and managing **100,000+ ratings** from real-world datasets, improving data handling accuracy by **40%**.

Dynamic Memory Allocator

- Developed a high-performance memory allocator in **C/C++**, optimizing memory management for real-time system demands, crucial for efficient model deployment in autonomous systems.
- Debugged and optimized the allocator using **GDB**, identifying and resolving memory leaks and fragmentation issues to ensure optimal performance.
- Engineered direct **OS** memory requests for large allocations, optimizing memory utilization by over **15%** for allocations exceeding pool capacities.

AFFILIATIONS

Junior Representative for National Society of Black Engineers (**NSBE**), Louis Stokes Alliance for Minority Participation (**LSAMP**), Collegiate Science and Technology Entry Program (**CSTEP**), ColorStack, CodePath